

A Guide to the Essential Protocols for the IoT

The IoT extends from a single constrained device to a whole range of cloud systems, all connected by a set of IoT protocols that allow devices and servers to talk to one another.

When you surf the Internet or purchase a cellphone on, say, Flipkart, you use Hyper Text Transfer Protocol Secure (HTTPS), Transmission Control Protocol (TCP), Domain Name System (DNS), IPv6 or IPv4 data link layer protocols and 802.3 protocols. These are basic networking protocols inside a browser and the computer's operating system and asymmetric digital subscriber line (ADSL) router. You use these protocols even while checking an email on your computer.

Protocols are the building blocks of all networking technologies on the Internet. These are software, and the size of these protocols in Windows operating system is in megabytes. Researchers all over the world are trying to reduce the size of TCP/IP protocols.

HTTP is a software protocol, which comes inbuilt in the Internet browser, such as Chrome. It sends commands to the Web server and gets replies from the same. This software listens on port number 80 and receives IP packet on 10.100.10.11 IP address.

Let us take an example. Flipkart uses the secure HTTPS protocol to communicate with the browser. This is a new protocol. It listens on port number 443 and 10.100.10.11 IP address. (Port number is a 16-bit logical number between 0 and 65535.) Here, IP address did not change, but port number did.

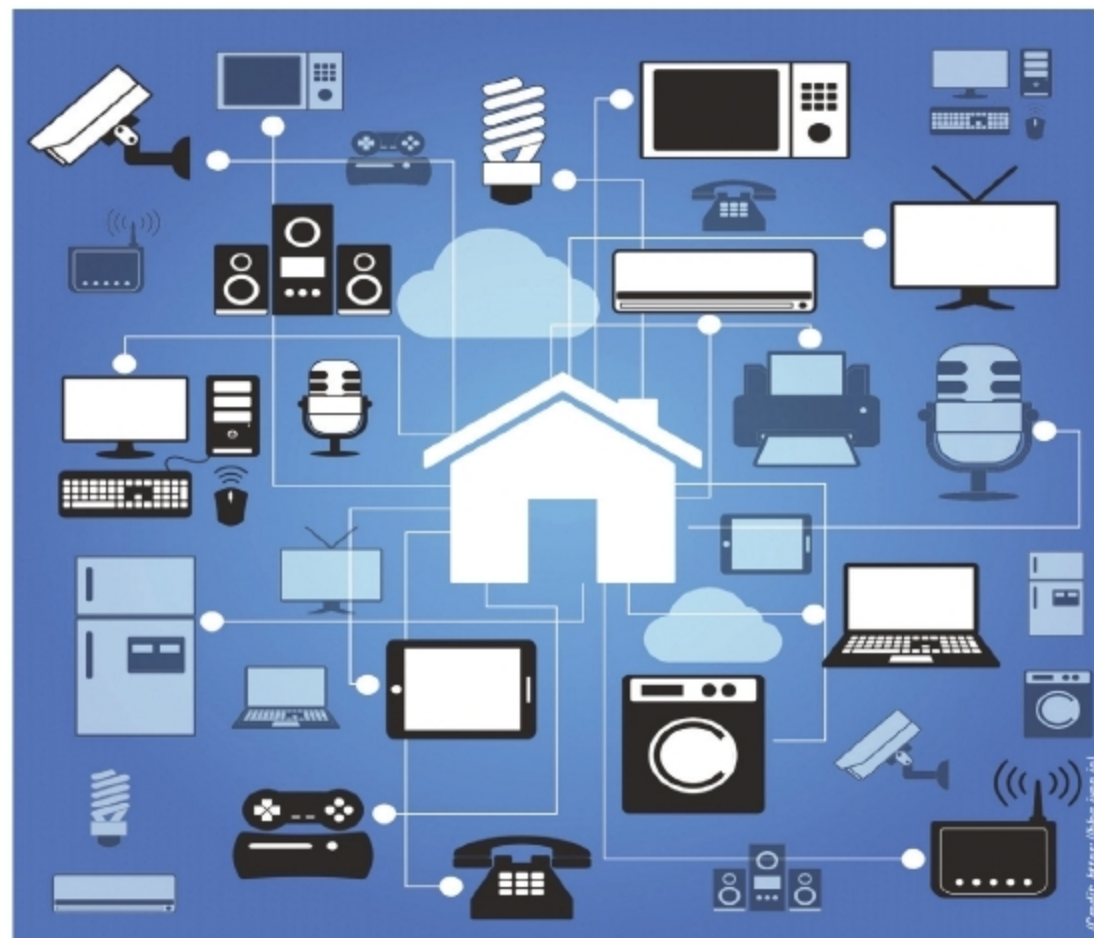
Port number and IP address form the socket. Hence, the Internet uses socket for communication. A secure socket opens between the browser and Flipkart's Web server. DNS protocol resolves *www.flipkart.com* string to the IP address of Flipkart's Web server. This IP address is fixed.

TCP protocol helps carry packets from the computer to Flipkart's server, bit by bit without error. Ethernet 802.3 protocol provides packets at the local network on the physical layer of Flipkart's office. Flipkart's server must be connected to a LAN or optical network.

This infrastructure is available for understanding the

communication between two computers. This is how we use the Internet or shop online on e-commerce websites such as Flipkart.

Now, let us see how we can relate the above narrative to an Internet of Things (IoT) system. Basic functions of Internet communication remain the same. An IoT system also uses HTTP protocol. This protocol uses GET/POST commands to communicate with the Web server. But instead of the browser sending commands to the server, the IoT device



(hardware browser) sends commands automatically to a pre-defined IP address of the manufacturer's server.

HTTP protocol is programmed in a tiny microcontroller (MCU). The MCU is part of the IoT sensor/device.

Similarly, other protocols are also used in the IoT ecosystem. However, focus of the IoT device is to design a constrain-oriented system. IoT devices work on wired or wireless networks, with low power and small memory footprint requirements. Hence, many new protocols are evolving for the IoT, and older ones are being adapted as per the functionalities of the IoT.

Functions of an IoT system are given below:

- Addressing constrained IoT devices/sensors
- Handling packet delivery from IoT devices to gateway, server or cloud